cosmic flow, such excess motion is minimal, fully in keeping with special relativity, even though their motion relative to each other, arising from the swelling of space itself, may exceed the speed of light.[20]

Third, if space is expanding, wouldn't that mean that in addition to galaxies being driven away from each other, the swelling space within each galaxy would drive all its stars to move farther apart, and the swelling space within each star, and within each planet, and within you and me and everything else, would drive all the constituent atoms to move farther apart, and the swelling of space within each atom would drive all the subatomic constituents to move farther apart? In short, wouldn't swelling space cause *everything* to grow in size, including our meter sticks, and in that way make it impossible to discern that any expansion had actually happened? The answer: no. Think again about the balloon-and-penny model. As the surface of the balloon swells, all the pennies are driven apart, but the pennies themselves surely do not expand. Of course, had we represented the galaxies by little circles drawn on the balloon with a black marker, then indeed, as the balloon grew in size the little circles would grow as well. But pennies, not blackened circles, capture what really happens. Each penny stays fixed in size because the forces holding its zinc and copper atoms together are far stronger than the outward pull of the expanding balloon to which it is glued. Similarly, the nuclear force holding individual atoms together, and the electromagnetic force holding your bones and skin together, and the gravitational force holding planets and stars intact and bound together in galaxies, are stronger than the outward swelling of space, and so none of these objects expands. Only on the largest of scales, on scales much larger than individual galaxies, does the swelling of space meet little or no resistance (the gravitational pull between widely separated galaxies is comparatively small, because of the large separations involved) and so only on such super galactic scales does the swelling of space drive objects apart.

## Cosmology, Symmetry, and the Shape of Space

If someone were to wake you in the middle of the night from a deep sleep and demand you tell them the shape of the universe—the overall